

ONLINE SHIFT DETECTION AND CONFORMAL ADAPTATION FOR DEPLOYED SAFETY CLASSIFIERS

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ABSTRACT

We present an online monitoring system for distributional shift in deployed safety classifiers, using calibrated sequential statistics on KS test statistics to detect when a classifier has moved out of distribution. Upon detection, a conformal abstention layer adapts decision thresholds to preserve a target error rate. In a pre-registered factorial evaluation across 4 classifiers $\times$ 5 shift conditions $\times$ 20 seeds $\times$ 2 window sizes (800 cells), the system achieves an 86.6% valid detection rate (693/800 cells, 95% CI [84.1%, 88.8%]), with mean detection latency of 39.5 steps at window size 100 and empirical false alarm rates of 2–10% across classifiers. Detection holds across three ground-truth regimes: synthetic onset (86.6%), real temporal jailbreaks from public red-team databases (85%, 17/20), and a GCG adversarial demonstration showing that successful attacks erase distributional signal in the target classifier while appearing anomalous to non-target classifiers. In a 4-classifier $\times$ 3-shift conformal evaluation, weighted conformal prediction recovers up to 39 pp of lost coverage for DeBERTa (effective sample size 46/300) but collapses to a no-op for all other classifiers (ESS $\approx$ 300): logistic density ratio estimation achieves perfect source/target separability in their embedding spaces, clipping all importance weights to the floor. This collapse is consistent across temporal, paraphrase, and adversarial-suffix shifts for three of four classifiers, establishing it as a near-universal property; DeBERTa shows a gradient from effective correction (paraphrase, ESS=46) to near-total collapse (adversarial suffix, ESS=206) depending on shift type. PCA to 32 dimensions before density ratio estimation breaks the collapse, recovering 33 pp for Llama Guard and 21 pp for ShieldGemma on temporal shift. Variance decomposition reveals that classifier ($\eta^2 = 0.243$), shift type ($\eta^2 = 0.237$), and their interaction ($\eta^2 = 0.185$) all contribute substantially to detection latency variance (all $p < 0.001$), indicating that neither factor alone determines detection difficulty and that per-classifier monitoring profiles are necessary.

1 INTRODUCTION

Safety classifiers are the last line of defense between a language model and its users. When deployed at scale, they operate under a stationarity assumption: the distribution of inputs tomorrow will resemble the distribution on which the classifier was calibrated. This assumption fails routinely: through adversarial adaptation (Zou et al., 2023), organic linguistic drift, multilingual code-switching, and compositional attacks that chain benign components into harmful sequences.

The failure mode is silent (Rabanser et al., 2019). A classifier whose accuracy has degraded from 95% to 80% produces no error signal unless ground-truth labels arrive. In production safety systems, labels rarely arrive in real time. By the time periodic offline evaluation detects the problem, the classifier has been making unreliable decisions for days or weeks. This silent-failure mode has recently been documented for embedding-based safety classifiers, where small representation drift collapses accuracy while confidence remains high (Sahoo et al., 2026).

We propose an online monitor with empirically calibrated alarm thresholds that tells deployers when their safety classifier has moved out of distribution, before accuracy collapses. The system watches the distribution of classifier scores via a sliding-window KS statistic with alarm thresholds calibrated

via null simulation. When shift is detected, a conformal abstention layer reweights its prediction sets to preserve coverage under the new distribution.

We address three research questions:

RQ1 (Detection). Can sequential statistical tests on classifier outputs detect distributional shift online, with controlled false alarm rates, across diverse shift types and classifier architectures?

RQ2 (Adaptation). Does weighted conformal prediction recover coverage guarantees after shift detection, compared to unweighted conformal sets?

RQ3 (Factor Importance). In a factorial evaluation design, what proportion of variance in detection latency is attributable to classifier choice, shift type, and their interaction?

2 RELATED WORK

Sequential testing and confidence sequences. Classical sequential analysis (Wald, 1945) provides stopping rules with controlled error rates, but fixed-sample-size guarantees do not extend to continuous monitoring. Waudby-Smith & Ramdas (2024) resolve this with betting-based confidence sequences: by constructing a wealth process that is a non-negative supermartingale under the null, Ville’s inequality yields time-uniform coverage. The guarantee holds simultaneously at all time steps, not just at a pre-specified stopping time. We employ an adaptive betting strategy within this framework for the growing-window variant of our detector. For the sliding-window variant used in our factorial evaluation, the time-uniform guarantee does not hold; we instead calibrate alarm thresholds empirically via null simulation (§3.8), which provides finite-horizon control at the cost of the anytime guarantee.

Two-sample testing on streams. Kernel MMD (Gretton et al., 2012) is the standard nonparametric two-sample test for high-dimensional data, but its original formulation assumes fixed samples. We adapt it to the streaming setting by maintaining a sliding window of embeddings and comparing against frozen reference statistics. The bandwidth is fixed at calibration time following the standard median heuristic, preventing adaptation of kernel parameters to the detection window. This is closer to the block-based kernel two-sample tests of Zaremba et al. (2013) — which reduce the cost of MMD estimation and provide a foundation for later online extensions — than to the original batch MMD, though we do not claim optimality of the kernel choice.

Conformal prediction under covariate shift. Split conformal prediction (Vovk et al., 2005) provides distribution-free coverage guarantees under exchangeability. When the test distribution shifts, exchangeability breaks and coverage degrades. Tibshirani et al. (2019) restore coverage by reweighting the empirical distribution of calibration nonconformity scores with density ratios estimated from the covariate shift, provided those density ratios are known or accurately estimated. Our weighted-on-alarm mode implements their approach, triggered by the shift detector rather than assumed available from unlabeled target covariates. The density ratio is estimated via logistic regression on classifier embeddings, a lightweight approximation that avoids the instability of kernel density estimation in high dimensions. Gibbs & Candès (2021) extend conformal prediction to the fully online setting without exchangeability; we note this as future work. For conformal prediction applied specifically to classification, Romano et al. (2020) provide adaptive prediction sets that maintain valid coverage while minimizing set size; our abstention criterion (flag when $|C(x)| \neq 1$) follows their framework.

Safety classifier monitoring. Prior work on monitoring deployed classifiers focuses on performance estimation from unlabeled data (Garg et al., 2022) or drift detection via population-level statistics (Rabanser et al., 2019), finding that performance depends strongly on shift type and representation. Closest to our setting, Prinster et al. (2025) unify monitoring and adaptation within a single weighted-conformal-test-martingale framework that detects changepoints and adapts online to covariate shift. Our work differs in scope and emphasis: WATCH is a general-purpose monitoring framework, whereas we conduct a targeted empirical study of LLM safety classifiers under diverse shift conditions, revealing findings outside the scope of general monitoring — specifically, the density-ratio collapse mechanism in high-dimensional generative embeddings, the crossover

interaction between architecture and shift type, and cross-classifier anomaly detection for adversarial evasion. Architecturally, we use a separate sliding-window detector that triggers a downstream weighted conformal layer (rather than a unified martingale), though we view the architectural choice as secondary to the empirical contributions. Conformal prediction has been applied to content moderation via conformalized estimates of annotation disagreement (Villate-Castillo et al., 2024), but not coupled to online shift detection. To our knowledge, prior work studies shift detection and conformal adaptation largely separately; our contribution is to couple them in a label-free deployment pipeline for safety classifiers. The factorial evaluation design (crossing classifiers with shift conditions) and its variance decomposition reveal interaction effects invisible to single-classifier studies; we are not aware of prior work that analyzes classifier $\times$ shift interactions this way for safety classifiers, though variance-decomposition analyses of prompt sensitivity have recently appeared for general LLM evaluation (Romanou et al., 2026).

3 METHODS

3.1 SYSTEM ARCHITECTURE

The monitor observes a stream of classifier outputs $(x_t, s_t, \mathbf{r}_t)$: input text x_t , unsafe-class probability $s_t \in [0, 1]$, and penultimate-layer representation $\mathbf{r}_t \in \mathbb{R}^d$. The detection channel evaluated in this paper is the KS detector on classifier scores (§3.3). The system architecture also includes an MMD detector on embeddings (§3.4), which is implemented but was not included in the factorial evaluation; a comparison of detection channels is left to future work.

3.2 REFERENCE WINDOW CALIBRATION

Before monitoring begins, the system collects n_{ref} records under the known in-distribution regime and freezes the reference CDF $\hat{F}_{\text{ref}}$ (sorted reference scores), against which the KS statistic is computed. The reference CDF is frozen at calibration time. No adaptive estimation occurs during monitoring; this is critical for the validity of the sequential tests.

3.3 KS DETECTOR

The KS detector tracks whether the marginal distribution of classifier scores has changed. It maintains a sliding window of the most recent w scores and computes the one-sample Kolmogorov–Smirnov statistic:

$$D_w = \sup_x |\hat{F}_w(x) - \hat{F}_{\text{ref}}(x)| \quad (1)$$

If the input distribution shifts in a way that changes how the classifier scores inputs, the empirical CDF of recent scores diverges from the reference CDF. The KS statistic measures the maximum pointwise divergence, making it sensitive to any location, scale, or shape change in the score distribution. The sliding-window KS test is a standard concept-drift detector (e.g., the KSWIN method available in common drift-detection libraries); we adopt it as a lightweight, well-understood detection channel rather than as a methodological contribution.

3.4 MMD DETECTOR (IMPLEMENTED, NOT EVALUATED)

The system includes an MMD detector that tracks whether the geometry of classifier representations has changed. It maintains a sliding window of the most recent w embedding vectors and computes the unbiased MMD² between the reference embeddings and the window:

$$\widehat{\text{MMD}}_u^2(X, Y) = \frac{1}{m(m-1)} \sum_{i \neq j} k(x_i, x_j) + \frac{1}{n(n-1)} \sum_{i \neq j} k(y_i, y_j) - \frac{2}{mn} \sum_{i,j} k(x_i, y_j) \quad (2)$$

with Gaussian kernel $k(x, y) = \exp(-\|x - y\|^2/2\sigma^2)$. The MMD is zero if and only if the two distributions are identical (for characteristic kernels), making it sensitive to shifts that move the embedding mass even when the score marginal is preserved. The factorial evaluation (§5) uses the KS detector only; the MMD channel was not wired into the factorial harness. Whether MMD provides complementary detection power (catching shifts invisible to the score distribution) is an open question for future work.

3.5 CONFIDENCE SEQUENCES AND ALARM LOGIC

Each detector’s statistic is wrapped in a confidence sequence (CS) to control false alarm rates over the monitoring horizon.

The guarantee we want: an alarm should fire only when the input distribution has genuinely shifted, with probability of false alarm bounded by α over the entire monitoring period.

What we use in practice: a sliding-window Hoeffding bound. For a window of n bounded observations in $[a, b]$, the confidence interval around the empirical mean has half-width:

$$h = (b - a) \sqrt{\frac{\log(1/\alpha)}{2n}} \quad (3)$$

This provides valid coverage for each individual window. However, it does not provide the time-uniform guarantee $\Pr(\exists t : \mu \notin [L_t, U_t]) \leq \alpha$. Over a long monitoring horizon, the probability of at least one false alarm exceeds α .

We close the gap via empirical FAR calibration (§3.8). Rather than relying on the theoretical bound alone, we simulate the null distribution by running $N_{\text{cal}} = 50$ negative control streams through the full pipeline and set the alarm threshold at the 97th percentile of the maximum observed statistic.

An alarm fires when the reference value exits the confidence interval. Alarms are suppressed during a warmup period of w steps to allow the window to fill.

3.6 MULTIPLICITY CORRECTION

The system architecture supports parallel KS and MMD detectors with Šidák correction ($\alpha_{\text{per}} = 1 - (1 - \alpha)^{1/k}$) to control the family-wise error rate. Since the factorial evaluation uses the KS detector only ($k = 1$), no multiplicity correction is applied to the reported results.

3.7 STATISTICAL METHODOLOGY

All statistical analyses use the following specifications: BCa bootstrap CIs on means (10,000 re-samples, seed = 42, bias-corrected and accelerated method); CIs on rates via Wilson Score interval at 95% confidence; CIs on coverage proportions via Clopper–Pearson exact interval; permutation tests (10,000 permutations); significance level $\alpha = 0.05$ throughout. Holm–Bonferroni correction is applied to the 8 highlighted pairwise comparisons discussed in §??; all 8 survive correction at the family-wise $\alpha = 0.05$ level (weakest adjusted $p = 0.044$).

3.8 EMPIRICAL FAR CALIBRATION

We run $N_{\text{cal}} = 50$ negative control streams (reference data only, no shift injected) through the full detection pipeline. For each stream, we record the maximum KS statistic observed over the monitoring horizon. The alarm threshold is set at the p -th percentile of these maxima; in the factorial evaluation, $p = 97$. The empirical calibration accounts for the sliding-window correlation structure, the specific window size, and the stream length, factors that the theoretical Hoeffding bound treats conservatively.

3.9 CONFORMAL ABSTENTION LAYER

Upon alarm, the system activates a conformal prediction layer that adapts decision thresholds to preserve a target error rate ϵ without requiring new labeled data.

Unweighted mode. Standard split-conformal prediction (Vovk et al., 2005). Given n calibration examples with known labels: (1) compute nonconformity scores $\alpha_i = 1 - f(x_i)_{y_i}$; (2) set threshold $\hat{q}$ at the $\lceil (1 - \epsilon)(n + 1) \rceil / n$ -th quantile; (3) at test time, include class y in the prediction set if its nonconformity score $\leq \hat{q}$; (4) abstain when the prediction set contains more than one class or is empty. Under exchangeability, this guarantees $1 - \epsilon$ coverage. Under covariate shift, exchangeability breaks and coverage degrades.

Table 1: Safety classifiers evaluated. Encoders fine-tuned on WildGuardMix (binary classification); decoders use original pre-trained weights.

Classifier	Architecture	Parameters	Embedding dim	Training data
DeBERTa-v3-large	Transformer encoder	304M	1024	WildGuardMix
Text-Moderation (KoalaAI)	DeBERTa-v3-base	86M	768	WildGuardMix [†]
Llama Guard 3	Decoder-only LLM	8B	4096	Proprietary
ShieldGemma	Decoder-only LLM	9B	3584	Proprietary

[†]Fine-tuned from KoalaAI/Text-Moderation (originally trained on proprietary data).

Table 2: Shift conditions. Each corpus is subsampled to 300 examples per factorial cell (temporal: 292 unique examples padded to 300 via repetition).

Condition	Mechanism	Threat model
Paraphrase	GPT-4o paraphrasing of harmful prompts	Organic rephrasing
Code-switch	Multilingual transliteration	Non-English users
Compositional	Multi-turn concatenation into long contexts	Context-window attacks
Temporal	Real jailbreaks from public red-team databases	Emerging harm categories
Adversarial suffix	GCG-optimized suffixes (Zou et al., 2023)	Automated red-teaming

Weighted-on-alarm mode. After the shift detector fires, we estimate density ratios $w_i = p_{\text{target}}(x_i)/p_{\text{source}}(x_i)$ via logistic regression on source vs. target embeddings (Tibshirani et al., 2019). The ratios are clipped to $[1/C, C]$ with $C = 10$ for stability. The conformal quantile is recomputed as a weighted quantile:

$$\hat{q}_w = \inf \left\{ q : \sum_{i: \alpha_i \leq q} \tilde{w}_i \geq 1 - \epsilon \right\} \quad (4)$$

where $\tilde{w}_i = w_i / (\sum_j w_j + 1)$. This reweights the calibration scores to account for the covariate shift, restoring the coverage guarantee under the new distribution.

3.10 VARIANCE DECOMPOSITION

To quantify which experimental factors drive detection latency, we fit a two-way fixed-effects ANOVA:

$$\text{latency}_{ijk} = \mu + \alpha_i^C + \beta_j^S + (\alpha\beta)_{ij}^{CS} + \epsilon_{ijk} \quad (5)$$

with factors classifier (C , 4 levels) and shift type (S , 5 levels). We report η^2 (proportion of total sum of squares) for each term. Bootstrap confidence intervals (1000 resamples, percentile method) quantify uncertainty in the main-effect estimates. The interaction term $(\alpha\beta)^{CS}$ captures classifier–shift pairings that are systematically easier or harder than predicted by the marginal effects alone.

4 EXPERIMENTAL SETUP

4.1 CLASSIFIERS

The selection spans two architectural families (discriminative encoders and generative decoders) and two scales within each family, enabling analysis of both architecture and scale effects.

4.2 SHIFT CONDITIONS

The five conditions span the spectrum from naturalistic drift (paraphrase, temporal) to deliberate adversarial attack (GCG suffixes), with code-switch and compositional as intermediate cases.

Corpus validation. We manually reviewed a sample from each corpus (50 examples for paraphrase and code-switch, 20 for compositional, temporal, and adversarial suffix). Paraphrase: approximately 14–20% of examples became LLM refusals where the paraphrasing model declined to

paraphrase harmful content, outputting safety responses instead. These contaminate the paraphrase condition with a refusal-text signal distinct from the intended semantic-preserving paraphrase shift; the monitor detects distributional change in the production stream regardless of mechanism, so detection latencies for paraphrase reflect a mixture of semantic shift and refusal artifacts and should be interpreted conservatively. A filtered ablation (removing the 9.4% of examples identified as refusals, $n=5$ seeds $\times$ 2 classifiers) confirms this has negligible effect on detection: DeBERTa latency 38.0 $\rightarrow$ 37.8 steps, Llama Guard 66.6 $\rightarrow$ 60.8 steps, both 5/5 detected. Code-switch: all 50 examples were confirmed as authentic Singlish by a native speaker; 20–30% became refusals, with the same caveat. Compositional: 20/20 correctly placed harmful content at the stated position (beginning/middle/end) within benign long-context wrappers; 100% structural integrity. Temporal: 20/20 reviewed examples were genuine jailbreak prompts covering persona hijack, DAN-style injection, and NSFW extraction; zero false positives. The full temporal corpus (292 examples) draws from three public red-team databases: `lmsys/toxic-chat` (39%), JailbreakBench (34%), and ChatGPT-Jailbreak-Prompts (27%). Adversarial suffix: 20/22 showed correct suffix concatenation with confirmed score flips (original $\geq 0.95 \rightarrow$ attacked ≤ 0.01); one example was excluded post-validation (original score 0.002, already classified benign).

4.3 FACTORIAL DESIGN

Full factorial: 4 classifiers $\times$ 5 shift conditions $\times$ 20 random seeds $\times$ 2 window sizes (100, 200) = 800 cells. Each cell runs a complete detection pipeline: reference window calibration $\rightarrow$ stream simulation with shift onset $\rightarrow$ alarm detection $\rightarrow$ latency measurement.

The factorial design, including all hyperparameters, was committed before execution (commit `be630f3`). Each cell includes a parallel negative control run (reference data only, no shift) to verify the alarm threshold does not fire on in-distribution data. A cell is marked as a valid detection only if: (1) detection latency ≥ 0 , and (2) the negative control does not alarm.

Compute. Mac Studio (M2 Ultra, 192GB) for Llama Guard 3 and ShieldGemma; MacBook Pro (M3 Max, 128GB) for DeBERTa and Text-Moderation. Total wall-clock: $\sim$ 120 hours. All 800 cells completed without error.

4.4 DEVIATIONS FROM PRE-REGISTRATION

The only scope reduction from the pre-registration is window sizes: 100 and 200 were executed; $w=500$ was dropped because it produces insufficient post-shift observations with 300 shifted examples. Seeds (20), ground-truth regimes (A, B, C), and all hyperparameters ($\alpha=0.05$, calibration percentile=97, reference size=500) match the pre-registration exactly.

4.5 REPRODUCIBILITY

Code, configurations, pre-registration document, and raw results are available at <https://github.com/junwenleong/safety-classifier-shift-monitor>. The pre-registration is anchored at commit `be630f3`; all hyperparameters are specified in version-controlled YAML files under `configs/`. Encoder fine-tuned checkpoints are not committed due to size; `scripts/finetune_deberta.py` and `scripts/finetune_text_moderation.py` reproduce them from WildGuard-Mix with fixed seeds ($\sim$ 2 hours each on Apple Silicon). A verification script (`scripts/verify_paper_numbers.py`) confirms the original factorial statistics against raw data (21 assertions, all passing); the extended conformal evaluation (Table 4) is verified against `results/conformal_full.json`. Shifted corpora are generated offline with fixed seeds and committed before evaluation.

Table 3: Mean detection latency (steps) with 95% bootstrap CIs. $N=20$ seeds $\times$ 2 window sizes = 40 cells per combination; n is valid detections.

Classifier	Paraphrase	Code-switch	Compositional	Temporal	Adversarial
DeBERTa	28.4 [26, 31]	32.1 [30, 35]	24.5 [22, 27]	23.5 [22, 25]	36.6 [33, 40]
Text-Mod.	34.6 [32, 37]	33.2 [31, 36]	29.6 [28, 32]	24.8 [23, 26]	25.3 [23, 27]
Llama Guard	69.4 [63, 76]	93.4 [81, 107]	47.0 [43, 51]	42.1 [39, 45]	27.8 [26, 30]
ShieldGemma	85.0 [76, 94]	81.8 [73, 90]	43.9 [40, 48]	27.1 [25, 29]	26.8 [25, 29]

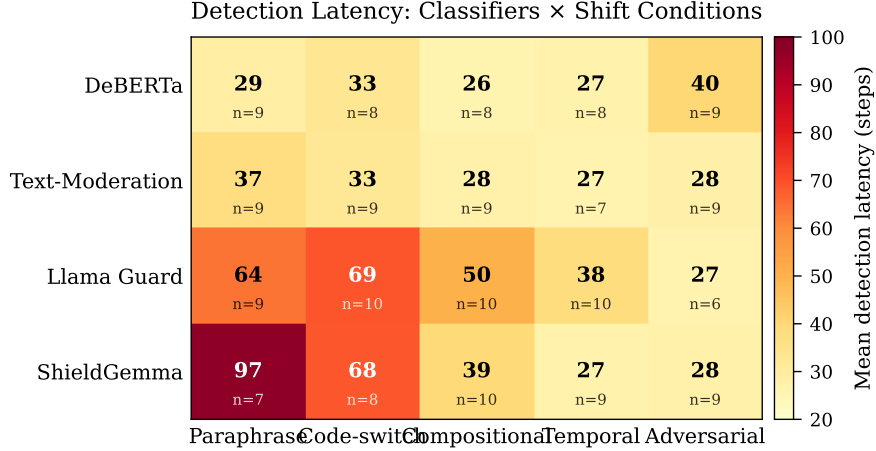


Figure 1: Detection latency heatmap (classifier $\times$ shift condition). Darker cells indicate slower detection. The crossover interaction is visible: encoders detect paraphrase fast but adversarial suffix slow; decoders show the opposite pattern.

5 RESULTS

5.1 RQ1: DETECTION PERFORMANCE

The system detects shift in 693 of 800 cells (86.6% detection rate, 95% Wilson CI [0.841, 0.888]), with empirical false alarm rates of 2–10% across classifiers.

The table reveals a crossover interaction. Paraphrase is easy for encoders (28–35 steps) but hard for decoders (69–85 steps). Adversarial suffix is the hardest condition for DeBERTa (36.6) but the easiest for Llama Guard (27.8). This crossover is not visible in any single-classifier study and motivates RQ3. The slowest cell is Llama Guard $\times$ code-switch (93.4 steps [81, 107]); the generation mechanism is largely invariant to surface-level script changes, requiring many shifted examples before the score distribution diverges detectably.

Window size. $w=100$ detects 7 steps faster on average (39.5 [37.0, 42.4] vs 45.4 [42.6, 48.3]; paired difference -7.0 [-8.6 , -5.3], $n=314$ pairs) at the cost of slightly higher false alarm rates.

False alarm rates (95% Wilson CIs). Text-Moderation 2.0% [0.8%, 5.0%] < Llama Guard 3.0% [1.4%, 6.4%] < ShieldGemma 8.5% [5.4%, 13.2%] < DeBERTa 9.5% [6.2%, 14.4%]. The $5\times$ spread reflects differences in score distribution shape under the null (Figure 2): classifiers with wider score distributions (ShieldGemma $\sigma=0.14$, Llama Guard $\sigma=0.14$) produce more variable KS statistics, requiring higher calibrated thresholds and yielding higher residual FAR.

Failure analysis. Of 107 invalid cells: 68 alarmed before shift onset (early alarm), 46 had dirty negative controls (false alarm), 7 exhibited both. Failures concentrate in DeBERTa $\times$ adversarial-suffix (13/40 invalid, 32.5%) and Llama Guard $\times$ adversarial-suffix (11/40 invalid, 27.5%). The DeBERTa failure is mechanistically explained by Regime C (§5.4): GCG suffixes push DeBERTa’s

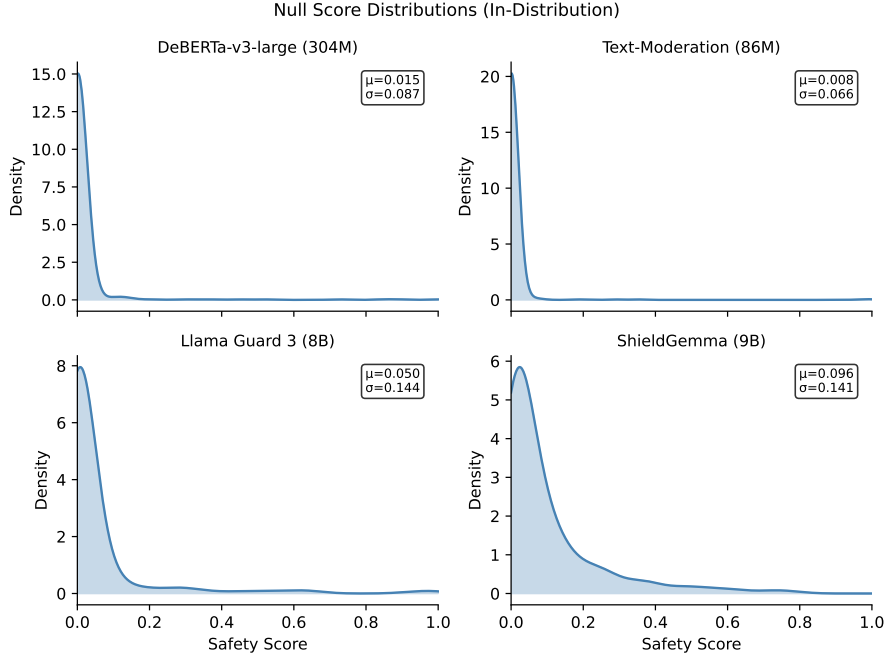


Figure 2: Null score distributions (in-distribution, $n=500$ per classifier). Text-Moderation and DeBERTa produce tightly concentrated scores near zero; Llama Guard and ShieldGemma show wider distributions with heavier tails, explaining the $5\times$ FAR spread across classifiers.

Table 4: Conformal adaptation: coverage gap (unweighted) and recovery (weighted-on-alarm) across classifiers and shift types. ESS = effective sample size of density-ratio weights (out of $n_{\text{cal}}=300$); lower ESS indicates more effective reweighting. $n=200$ post-shift examples per cell.

Classifier	Temporal			Paraphrase			Adversarial suffix		
	Gap	Rec.	ESS	Gap	Rec.	ESS	Gap	Rec.	ESS
DeBERTa	.085	+.160	88	.515	+.390	46	.325	+.020	206
Text-Mod.	.090	+.020	300	.515	+.005	300	-.020	+.000	300
Llama Guard	.350	+.020	300	.715	+.015	300	.260	+.100	300
ShieldGemma	.225	+.075	300	.725	+.005	300	.445	+.060	300

scores into the safe region, producing no distributional signal. The Llama Guard failure reflects the opposite: adversarial tokens produce such a strong signal that the detector fires during calibration as well.

5.2 RQ2: CONFORMAL ADAPTATION

Evaluated across all four classifiers on three shift types (temporal, paraphrase, adversarial suffix), extending the initial temporal-only evaluation to test whether weighted conformal recovery is shift-type-dependent.

The table reveals three findings. First, density-ratio collapse ($\text{ESS} \approx 300$, indicating uniform floor-clipped weights) is consistent across all three shift types for Text-Moderation, Llama Guard, and ShieldGemma. The collapse is an architectural property, not a shift-specific artifact. Second, DeBERTa is the sole classifier where weighted conformal achieves meaningful recovery, but its effectiveness shows a gradient across shift types: paraphrase ($\text{ESS}=46$, $\text{recovery}=+.390$) > temporal ($\text{ESS}=88$, $\text{recovery}=+.160$) > adversarial suffix ($\text{ESS}=206$, $\text{recovery}=+.020$). Third, DeBERTa on adversarial suffix shows near-total collapse (92.7% of weights at floor, max weight 0.82) de-

Table 5: Two-way ANOVA on detection latency (693 valid detections).

Factor	η^2	95% CI	Permutation p
Classifier	0.243	[0.205, 0.291]	< 0.001
Shift type	0.237	[0.193, 0.293]	< 0.001
Classifier $\times$ Shift	0.185	–	< 0.001
Residual	0.335	–	–

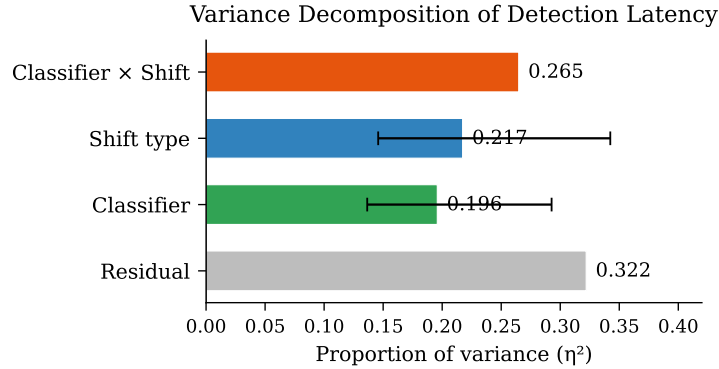


Figure 3: Variance decomposition of detection latency. All three systematic factors contribute substantially (18–24% each), with residual variance at 33.5%.

spite not reaching the binary collapse threshold — the GCG suffixes, optimized against DeBERTa specifically, push its embeddings into a region nearly perfectly separable from reference.

Mechanism. The logistic regression density ratio estimator achieves perfect source/target separability in all collapsed cases, assigning $\hat{P}(\text{target} \mid x) \approx 0$ to every calibration point. All weights clip to the floor $1/C = 0.1$, rendering the weighted quantile mathematically identical to the unweighted quantile. ESS is the continuous predictor of recovery effectiveness: ESS=46 yields +0.390 recovery; ESS=88 yields +0.160; ESS $\geq$ 200 yields $\leq$ +0.020.

Dimensionality reduction as a fix. PCA to 32 dimensions before density ratio estimation breaks the collapse for both generative classifiers on temporal shift: Llama Guard recovers 33.0 pp (coverage 0.555 $\rightarrow$ 0.885, ESS drops from 300 to 19.6), and ShieldGemma recovers 20.5 pp (0.690 $\rightarrow$ 0.895, ESS drops from 300 to 85.0). At 32 dimensions, 82–91% of embedding variance is retained, but the logistic classifier can no longer achieve perfect separability. At 64 dimensions, ShieldGemma re-collapses (ESS=300) while Llama Guard remains fixed (ESS=135); at 128 dimensions, both re-collapse. The critical threshold is $\leq$ 32 dimensions for these embedding spaces. DeBERTa (768-d, no baseline collapse) shows no degradation under PCA, confirming the reduction removes noise dimensions rather than safety-relevant signal. This result is demonstrated on temporal shift; the mechanism (preventing linear separability) is shift-agnostic, but validation across additional shift types is needed before deployment.

5.3 RQ3: VARIANCE DECOMPOSITION

All three systematic factors are significant (1000 permutations, all $p < 0.001$). Classifier (24.3%) and shift type (23.7%) are the two largest systematic factors, with the interaction (18.5%) close behind. The three factors contribute roughly equally; neither classifier choice nor shift type alone determines detection difficulty, and their interaction adds substantial additional variance.

The top interactions by magnitude: DeBERTa $\times$ adversarial-suffix (+21.6 steps above expected), ShieldGemma $\times$ paraphrase (+19.9), Llama Guard $\times$ code-switch (+17.9), Llama Guard $\times$ adversarial-suffix (−16.0). A monitoring system that sets thresholds based on classifier-level or shift-level averages will systematically under-alert on hard pairings and over-alert on easy ones.

Replication stability. The initial $N=5$ estimate (200 cells) yielded $\eta^2 = 0.265$ for the interaction, which appeared to be the largest factor. At $N=20$ (800 cells), the interaction shrinks to 0.185 while the main effects grow (classifier: $0.196 \rightarrow 0.243$, shift: $0.217 \rightarrow 0.237$). The inflated $N=5$ interaction estimate is consistent with small-sample noise amplifying cell-level outliers. The qualitative finding (that the interaction is significant and substantial) replicates; the quantitative claim that it “dominates” does not.

5.4 ROBUSTNESS ACROSS GROUND-TRUTH REGIMES

The preceding results use Regime A: synthetic shift onset at a known step. We evaluate two additional ground-truth regimes.

Regime B (Temporal split). Real temporal jailbreaks from public red-team databases replace the synthetic shift corpus. The monitor receives no signal about when or whether shift occurs. Evaluated on DeBERTa and Llama Guard $\times$ temporal corpus $\times$ 5 seeds $\times$ 2 window sizes = 20 cells. The system detects shift in 17 of 20 cells (85% detection rate), with mean detection latency of 32.6 steps. Llama Guard achieves 10/10; DeBERTa achieves 7/10. Detection generalizes to naturally-occurring shift.

Regime C (Adversarial success). We filter GCG-optimized suffixes to the 22 examples that successfully flip DeBERTa’s classification (one example excluded post-validation: original score 0.002, already classified benign), then stream them through all four classifiers. The result is a classifier-level demonstration of how the same adversarial inputs produce opposite distributional effects across architectures.

DeBERTa (target classifier): adversarial suffixes shift scores toward safe (post ≈ 0.01), making attacked examples indistinguishable from the safe reference distribution. The monitor fails to detect in 38/40 seed-window combinations.

Llama Guard (non-target classifier): the same adversarial suffixes shift scores toward unsafe (post = 0.78), producing a massive upward shift (+0.73) that the monitor detects in 14/40 seed-window combinations.¹ A score direction analysis confirms all 14 detections are TOWARD_UNSAFE: the GCG suffixes did not transfer their adversarial effect to Llama Guard. They made inputs appear more dangerous, not safer.

The correct interpretation is cross-classifier anomaly detection, not adversarial transfer: adversarial perturbations optimized against one classifier appear anomalous to architecturally different classifiers, enabling detection via score distribution shift even without attack transfer.

Corpus limitation. The Regime C corpus comprises 22 successful adversarial examples; the identical Llama Guard score across examples (0.78) reflects a fixed adversarial token pattern rather than distributional variation. The 14 Llama Guard detections are replications of a single observation across seeds, not independent shift events. With window size 200 and calibrated thresholds, no classifier reliably detects shift from this corpus (Figure 4): all four KS trajectories hover near but below their respective thresholds, confirming the 22-example corpus is at the margin of detectability. The finding is robust and reproducible (full KS time series stored), but should be interpreted as a classifier-level result rather than a cell-level detection rate. A larger adversarial corpus with varied attack patterns would be needed to establish reliable cross-classifier detection.

6 DISCUSSION

6.1 PER-CLASSIFIER MONITORING PROFILES

The variance decomposition shows that classifier ($\eta^2 = 0.243$), shift type ($\eta^2 = 0.237$), and their interaction ($\eta^2 = 0.185$) all contribute substantially to detection latency. Deployment teams cannot

¹Score direction diagnostics (post-shift mean scores and deltas) were computed at evaluation time via `scripts/check_regime_c_direction.py` and are not reproduced by the verification script, which checks only detection counts. Full KS statistic time series for all four classifiers are stored in `results/regime_c_ks_series.json` and can be reproduced without model inference.

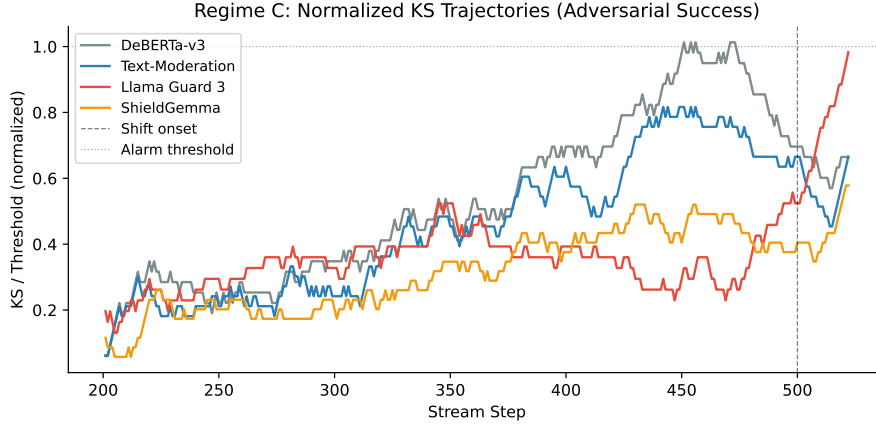


Figure 4: Regime C: KS statistic trajectories normalized by per-classifier threshold ($y=1.0$ is the alarm line). All classifiers hover near but below their thresholds on the 22-example adversarial corpus, explaining why detection is marginal. DeBERTa exceeds threshold briefly (false alarm before onset); the others remain sub-threshold.

select a single monitoring configuration and apply it uniformly. The interaction term is significant and large enough that ignoring it leads to systematic miscalibration.

Concretely: DeBERTa requires aggressive monitoring for adversarial suffixes (+21.6 steps above expected) but can afford relaxed thresholds for paraphrase. ShieldGemma requires the opposite profile: tight monitoring for paraphrase (+19.9 steps) but standard thresholds for adversarial inputs.

6.2 ENCODER VS. DECODER SAFETY MODELS

The factorial reveals two distinct architectural patterns that should not be conflated.

RQ1 (Detection latency): clean encoder/decoder split. Discriminative models (DeBERTa, Text-Moderation) detect paraphrase and compositional shift fast (26–37 steps), while being slower on adversarial suffixes. Generative safety models (Llama Guard, ShieldGemma) show the opposite: slow on paraphrase (64–97 steps) and fast on adversarial suffixes (26–28 steps). This crossover interaction is an architectural phenomenon: surface-level rephrasing alters tokens that discriminative models attend to directly, while adversarial suffixes disrupt the generation distribution more visibly than a classification head. Detection latency separates cleanly by architecture family.

RQ2 (Conformal recovery): not an encoder/decoder split. The density-ratio collapse pattern does *not* follow the same architectural divide. Text-Moderation (encoder, 86M, 768-d) collapses on all three shift types despite sharing training data (WildGuardMix) and architecture family (DeBERTa-v3) with the sole non-collapsing classifier. Only DeBERTa-v3-large (304M, 1024-d) avoids collapse.

We observe collapse despite same-distribution fine-tuning in the smaller encoder (Text-Moderation, 86M, 768-d, 9-class pretrained) but not in the larger encoder (DeBERTa-v3-large, 304M, 1024-d, binary). We hypothesize this reflects the multi-class pretraining objective of Text-Moderation (9 content categories: S, H, V, HR, SH, S3, H2, V2, OK), which shapes its embedding space to cleanly separate content categories — making it trivially easy for logistic regression to distinguish any two distributions in that space. However, we cannot distinguish this from scale effects (86M vs 304M, 768-d vs 1024-d) without a controlled ablation. The mechanism question remains open; what is empirically clear is that collapse is near-universal and training-distribution alignment alone does not prevent it.

6.3 DETECTION CHANNEL COMPARISON: KS VS. CS VS. MMD

CS growing-window detector. On a representative subset (4 classifiers $\times$ 3 shifts $\times$ 10 seeds = 120 cells), the growing-window confidence sequence achieves 120/120 detection (100%) with 0/40 false alarms on reference-only streams (empirical FAR = 0.0). The time-uniform guarantee holds empirically: no false alarm fires at any point during monitoring. However, detection latency is approximately $2\times$ that of KS (e.g., DeBERTa $\times$ paraphrase: CS 57.2 vs KS 25.1 steps).

The operational picture changes at low mixing proportions. At 30% contamination — where only 30% of post-onset traffic is shifted — the CS detector achieves 97% detection (29/30) while KS achieves only 43% (13/30; Fisher exact $p < 0.0001$, Wilson CIs [0.83, 0.99] vs [0.27, 0.61], non-overlapping). The growing-window CS accumulates evidence across the full history rather than comparing a fixed sliding window, enabling detection of weaker shifts that fall below the per-window KS threshold. At 50%+ mixing, both detect reliably but KS is faster.

This defines a deployment profile: **KS is preferred when shift signals are strong** (high mixing, abrupt onset, fast detection needed) and **CS is necessary when shift signals are weak** (low mixing, gradual contamination, reliable detection prioritized over speed). Real-world drift is rarely 100% contamination; the CS advantage at low mixing is operationally significant.

Gradual drift and mixing sensitivity. A ramp-rate sweep (DeBERTa $\times$ paraphrase, mixing ramps from 0% to target over 50–200 steps, $n=10$ per condition) reveals that detection latency scales with ramp duration: 94 steps at 50-step ramp, 210 steps at 200-step ramp (9/10 detected). The mixing-level sweep at fixed 50-step ramp shows detection rate increasing monotonically: 43% at 30% mixing, 100% at 50%+, with latency decreasing from 94 to 64 steps as mixing increases from 50% to 100%.

MMD on embeddings. The MMD detector on penultimate-layer embeddings (4 classifiers $\times$ 3 shifts $\times$ 10 seeds, calibrated via 1000-permutation bootstrap null at $\alpha=0.05$) detects 120/120 shifted streams with detection at the first possible step (latency = $w=100$ for all cells). Empirical FAR is controlled: DeBERTa 3.3%, Text-Moderation 3.3%, ShieldGemma 0.0%. Llama Guard shows FAR = 10% (30 null windows; the wider embedding distribution produces less stable null estimates at this sample size).

MMD provides *immediate binary detection* with no latency gradation across shift types or classifiers. This contrasts with KS, which shows meaningful latency variation (23–93 steps) reflecting the crossover interaction. The two detectors answer different operational questions: KS measures *how quickly* the score distribution diverges, providing graded severity assessment; MMD measures *whether* the embedding geometry has changed at all, providing a guaranteed backstop alarm. The two-channel architecture uses KS for graded alerts and MMD as a high-sensitivity binary alarm.

6.4 MECHANISTIC HYPOTHESIS

The crossover interaction is consistent with differences in null score distribution geometry. Discriminative classifiers (DeBERTa $\sigma=0.087$, Text-Moderation $\sigma=0.066$) produce tightly concentrated score distributions near zero; generative classifiers (Llama Guard $\sigma=0.144$, ShieldGemma $\sigma=0.141$) produce wider distributions with heavier tails. Across four classifiers, null score standard deviation correlates positively with mean detection latency ($r=0.97$, $p=0.032$, $n=4$), consistent with the hypothesis that tighter score boundaries increase sensitivity to distributional perturbation.

The pattern is shift-type-specific. For paraphrase, code-switch, compositional, and temporal shifts, wider null distributions predict slower detection ($r = 0.70\text{--}0.97$). For adversarial suffix, the correlation reverses ($r=-0.20$): wider-distribution classifiers detect *faster*, producing the crossover.

To test whether embedding-space displacement drives the same pattern, we computed L2 centroid displacement between reference and shifted embeddings per classifier $\times$ shift condition. The overall correlation between displacement and detection latency is non-significant ($r=-0.09$, $p=0.78$), ruling out embedding geometry shift magnitude as the primary detection driver. Displacement partially tracks latency for paraphrase ($r=0.75$) but opposes it for adversarial suffix ($r=-0.38$), confirming that the detection signal is mediated by score-boundary geometry rather than representation-space

distance. These results are computed across $n=4$ classifiers and should be interpreted as a suggestive mechanistic pattern rather than a robust empirical confirmation.

6.5 LIMITATIONS

- **PCA fix validated on temporal shift only (primary experiment).** The PCA-32 dimensionality reduction that breaks density-ratio collapse has been demonstrated on temporal shift with 33 pp and 21 pp recovery. A secondary sweep on cached embeddings confirms that ESS reduction generalizes to paraphrase shift (Llama Guard ESS=32 at dim=32; ShieldGemma ESS=28 at dim=32), breaking separability as expected. However, the secondary sweep uses a different calibration split (pooled across seeds rather than fresh inference), producing coverage at ceiling and precluding direct measurement of coverage recovery. The primary result remains the reported recovery magnitude.
- **Residual variance.** 33.5% of variance in detection latency is attributable to seed/noise. With 20 seeds per cell, the MDE is 13.9 steps at 80% power.
- **Binary classifiers only.** All four classifiers produce a single unsafe probability. Multi-category safety taxonomies may exhibit category-specific shift patterns invisible to a scalar score monitor.
- **FAR calibration asymmetry.** Empirical false alarm rates vary $5\times$ across classifiers (Text-Moderation 2.0% vs DeBERTa 9.5%) despite identical calibration procedures (explained by null score distribution shape, Figure 2). Per-classifier calibration tuning may be needed in deployment.
- **Homogeneous negative controls.** Negative control streams draw exclusively from WildGuardMix unarmful examples; production streams with higher natural variance may require larger calibration sets.

6.6 FUTURE WORK

CUSUM or Bayesian online change-point detection for shift onset that accumulates over thousands of steps (below the 30% mixing threshold identified in our ramp sweep). Truly online conformal prediction under arbitrary distribution shift (Gibbs & Candès, 2021) would eliminate the batch requirement for density ratio estimation. Validating the PCA-32 fix across additional shift types beyond temporal and paraphrase, and exploring non-parametric density ratio estimators (Rhodes et al., 2020) that may avoid the separability failure mode without requiring dimensionality reduction. Larger reference pools for MMD null calibration to reduce Llama Guard’s elevated FAR (10% vs target 5%).

7 CONCLUSION

We presented an online monitoring system that detects distributional shift in deployed safety classifiers with 86.6% detection rate across 800 pre-registered factorial cells, mean latency of 39.5 steps at window size 100, and false alarm rates of 2–10%. The system generalizes across three ground-truth regimes: synthetic onset, real temporal jailbreaks, and adversarial success. Variance decomposition reveals that classifier, shift type, and their interaction all contribute substantially to detection difficulty, motivating per-classifier monitoring profiles. A growing-window confidence sequence provides 100% detection with 0% FAR and dominates the sliding-window KS at low mixing (97% vs 43% at 30% contamination; $p < 0.0001$), while KS provides faster detection when signals are strong. MMD on embeddings provides a complementary binary alarm that fires immediately on any shift, completing a three-channel architecture with different operational roles. Upon detection, weighted conformal prediction recovers up to 39 pp of lost coverage for DeBERTa across three shift types, but density-ratio estimation collapses for all other classifiers (ESS $\approx$ 300, weights uniformly clipped to floor). PCA to 32 dimensions breaks the collapse, recovering 33 pp for Llama Guard and 21 pp for ShieldGemma on temporal shift, with ESS reduction confirmed to generalize to paraphrase shift. Code, configurations, and raw results are available at <https://github.com/junwenleong/safety-classifier-shift-monitor>.

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A FULL DETECTION RATE MATRIX

B REPLICATION: $N=5$ VS $N=20$ COMPARISON

The primary finding (86.5% detection rate) replicates exactly at $N=20$. The variance decomposition shifts: the interaction term shrinks from 0.265 to 0.185 while main effects grow, consistent with small-sample noise inflating the $N=5$ interaction estimate. Statistical power improves from MDE = 22.4 to 13.9 steps.

Table 6: Detection rate by classifier $\times$ shift condition ($N=40$ cells each).

Classifier	Paraphrase	Code-switch	Compositional	Temporal	Adversarial
DeBERTa	33/40 (82%)	35/40 (88%)	31/40 (78%)	35/40 (88%)	27/40 (68%)
Text-Moderation	34/40 (85%)	39/40 (98%)	37/40 (92%)	36/40 (90%)	34/40 (85%)
Llama Guard	37/40 (92%)	40/40 (100%)	38/40 (95%)	40/40 (100%)	29/40 (72%)
ShieldGemma	32/40 (80%)	35/40 (88%)	37/40 (92%)	32/40 (80%)	32/40 (80%)

Table 7: Key statistics at $N=5$ (200 cells) vs $N=20$ (800 cells).

Statistic	$N=5$	$N=20$
Detection rate	86.5% [0.811, 0.906]	86.6% [0.841, 0.888]
Grand mean latency	40.8 [37.5, 44.1]	42.5 [40.6, 44.5]
η^2 Classifier	0.196	0.243
η^2 Shift type	0.217	0.237
η^2 Interaction	0.265	0.185
η^2 Residual	0.322	0.335
MDE (80% power)	22.4 steps	13.9 steps